

Timer Replacement PCB — Complete Documentation

Translation. This is the English translation of the manual. The authoritative version is the German original ([Deutsch](#)). Also available: [Français](#).

Project: Replacement of a defective/outdated timer control PCB with an in-house development using an ESP32-C3, OLED display, Wi-Fi configuration and a Shelly 1PM Mini Gen4 as the power switch.

Status/Date: 2026-07-13 · Template version v8 · Phase: verification before layout **Author:** Silvio (hardware/measurements) + Claude (analysis/design)

1. What the device does (functional description)

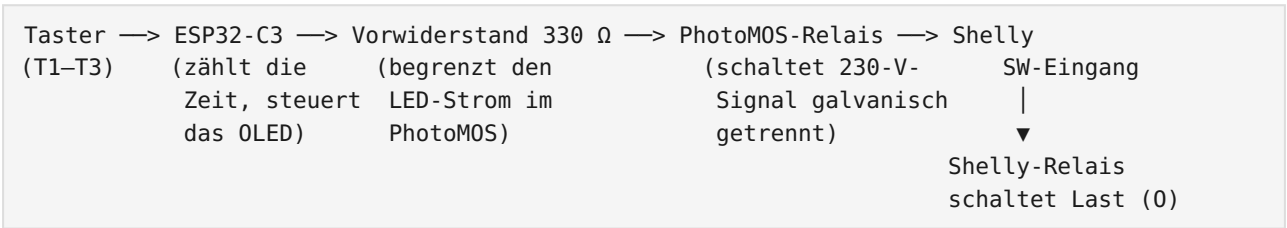
The device has three buttons on the front. Pressing a button switches a 230 V load on for an adjustable time:

Button	Default time	adjustable
T1	5 seconds	1-600 s via Wi-Fi
T2	10 seconds	1-600 s via Wi-Fi
T3	15 seconds	1-600 s via Wi-Fi

A small OLED display behind the viewing window shows the **Wi-Fi reception** (left) at the top, the large **time of day** (NTP, center) and the status "Ruhe" (Idle, right); at the bottom left the **date** (e.g. "Do 16.07.2026") and at the bottom right the **free memory**. While a timer is running, the center shows the **seconds countdown** and the right shows "Futter" (Feed). The three buttons (**S1 Down/Manual**, **S2 SET**, **S3 UP**) trigger timers, stop them and page through an info menu — details in Ch. 6.5.

Operation runs **without an app and without a cloud** via a small web page optimized for mobile devices, which the ESP32 serves itself on **port 80** (open `http://feeder-relais.local` in the browser): trigger buttons, set times, view network and status values. For integration (e.g. with ioBroker) the ESP additionally offers a lean **JSON API** (see Ch. 6.4). Optionally, the device can also be integrated into Home Assistant or ioBroker (esphome adapter).

1.1 The signal chain (this is how the signal flows)



Why a Shelly? The Shelly 1PM Mini Gen4 handles the actual switching of the load, measures the power in the process, and is additionally reachable via Wi-Fi/app. Our board only "presses" the switch for the Shelly — as a real hardware signal, **without a Wi-Fi dependency** for the core function.

What is a PhotoMOS? A tiny solid-state relay: on one side an LED (driven by the ESP with 3.3 V), on the other side a light-controlled switch that is allowed to switch mains voltage. Between them there is only light — i.e. a complete galvanic isolation between low voltage (SELV) and 230 V. The Shelly's SW input is mains-referenced (it expects a switched L), which is why a type with $\geq 400\text{ V}$ **reverse voltage is mandatory** (e.g. Omron G3VM-601BY or Panasonic AQY216). An ordinary optocoupler (PC817) is **not** permissible here.

The Shelly is configured to input mode "**Switch/Follow**": its relay is on for exactly as long as the ESP holds the signal. The timing is entirely up to the ESP.

2. Safety — read this first

- The device works with **230 V mains voltage. Danger to life!** Assembly, testing and commissioning belong in the hands of a suitably qualified person (a qualified electrician or an instructed person).
 - Initial functional tests of the ESP, buttons and OLED are performed **exclusively over USB** — with no mains voltage connected whatsoever (see the test plan in the project plan).
 - The first 230 V test is performed behind an **RCD/PRCD (portable residual current device plug)**, with the enclosure closed; never work on the open device while it is powered.
 - On the board the rule is: at least **6 mm of distance** (creepage distance) between all 230 V nets and the low voltage (SELV); details in Ch. 7.6.
 - A fuse (1 A slow-blow) and a varistor upstream of the power supply module are mandatory — the modules (TSP-05/HLK) do not bring either of them along themselves.
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3. Bill of materials (BOM)

Ref	Component	Value / type	Note
U1	ESP32-C3 Super Mini	18 × 24 mm, USB-C	controller, Wi-Fi
U2	PhotoMOS relay	Omron G3VM-601BY or Panasonic AQY216	$\geq 400\text{ V}$! SOP-4/DIP-4
PS1	AC/DC module	TENSTAR TSP-05 (5 V / 3 W)	HLK-PM05 clone, 34.7 × 20.5 × 15.05 mm, measure pin spacing!
—	Shelly 1PM Mini Gen4	S4SW-001P8EU	29 × 34 × 16 mm, max. 8 A/240 V
—	OLED display	0.91" SSD1306, 128 × 32, I2C	module 38 × 12 mm, pins: GND VCC SCL SDA, address 0x3C
SW1-3	Short-travel button	6 × 6 mm, travel 1.5 mm	pin pattern 6.9 × 4.4 mm (measured from the original)
R1	Resistor	330 Ω , axial	LED series resistor for PhotoMOS
F1	Fine-wire fuse	1 A slow-blow, 5 × 20 mm + holder	primary side
RV1	Varistor (MOV)	S14K275	primary side, in parallel
C1	Electrolytic cap	220 μF / 10 V	5 V buffer for Wi-Fi current peaks
C2	Ceramic cap	100 nF	5 V decoupling
J1	Mains terminal	2-pole (L, N)	screw terminal or solder pads
J2	Shelly connection	4 solder pads/pins	

Ref	Component	Value / type	Note
			L, N, SW, O — short wires to the Shelly screw terminals
J3	Load output	2-pole	O (switched), N
J4	OLED socket	socket header 1 × 4, pitch 2.54	build height per stack measurement (~11 mm), Ch. 5.6
—	Pin header 1 × 4	pitch 2.54	soldered to the OLED (facing backward)
—	Mounting material	3M VHB adhesive tape or 3D clip	fastening the Shelly + TSP

4. Enclosure and mechanics

4.1 The coordinate system (important for everything that follows!)

All dimensions refer to the **upper left corner of the ORIGINAL PCB**, x to the right, **y downward**, viewed from the **component side** (buttons at the top). The new, larger board extends beyond this origin — which is why there are **negative coordinates**. Advantage: all values ever measured remain valid forever, no matter how the contour grows.

4.2 Boundary conditions from the enclosure (measured/derived)

Quantity	Value	Source
Inner overall height	19.9 mm	measurement Silvio
Frame height (front panel)	2 mm	measurement Silvio
Space above the board	16.5 mm	measurement Silvio (board ↔ front)
Board thickness	1.6 mm	standard
Space below the board	≈ 1.8 mm	19.9 – 16.5 – 1.6 → nothing may protrude downward!
Height limit in the window area	≈ 12 mm	glass + OLED plugged in front of it
Board fastening	4 screws on 16.5 mm bosses on the front panel	with stabilizing ribs to the edge
Enclosure screwing	6 external screws	the board has 6 edge-open U-slots (width 10) instead of closed feed-throughs

4.3 Final board geometry (template v9)

All values are stored **machine-readable** in `hardware/platine_template.py` (the single source of truth!). Summary:

Element	Coordinates (mm)	Accuracy
Contour	x –14.2...91.1 / y –11.8...67.6 → 105.3 × 79.4 , corner radius 4 (locally reduced to 3.2 and 2.1 respectively at the outer boss slots)	±1 (left ±1.5)
4 mounting holes ø3.0	(3.8/4.1) (73.1/4.0) (3.7/39.6) (73.4/50.7)	±0.3

Element	Coordinates (mm)	Accuracy
6 boss slots (edge-open U, width 10)	upper row y −6.0 (opens toward the top edge), lower row y +64.0 (opens toward the bottom edge), row spacing 70; columns x −6.0 / 39.0 / 84.0 (pitch 45)	measured directly (Silvio 13.07.); absolute position still to be test-fitted
Buttons T1/T2/T3 (centers)	(13.2/12.3) (38.8/12.2) (64.1/12.0)	±0.3
Viewing window (reference)	x 18.4...53.6 / y 23.3...40.0	±1
J4 OLED socket (4 pins vertical)	pins at x = 15.0, y = 27.84 / 30.38 / 32.92 / 35.46	check assumptions, Ch. 5.6
Mounting area Shelly	x 56...90 / y 16.5...45.5 (horizontal)	—
Mounting area ESP32-C3	x 8.5...32.5 / y −10...8 (transverse, USB upward)	—
Mounting area TSP-05	x −1.5...33.2 / y 44.2...64.7 (AC side to the left)	—
Keepouts (reference)	ø8 around the 4 screw heads + 5 rib bands	rib directions = assumption!

Where do the values come from? Solder-side scan of the original PCB (300 dpi, de-rotated, mirrored) for buttons/holes/contour; enclosure scan, registered via the 4 frame screws (mirror transformation $x_{\text{platine}} = 110,5 - x_{\text{gehäuse}}$, $y_{\text{platine}} = y_{\text{gehäuse}} - 16,4$, ±0.4 mm); window position from the perspective-corrected photo.

4.4 The old cutout

The trapezoidal cutout on the bottom edge of the original PCB was the clearance for the middle lower screw boss (B5). In the new board it is **replaced by the edge-open U-slot B5** and is omitted.

5. Electronics in detail

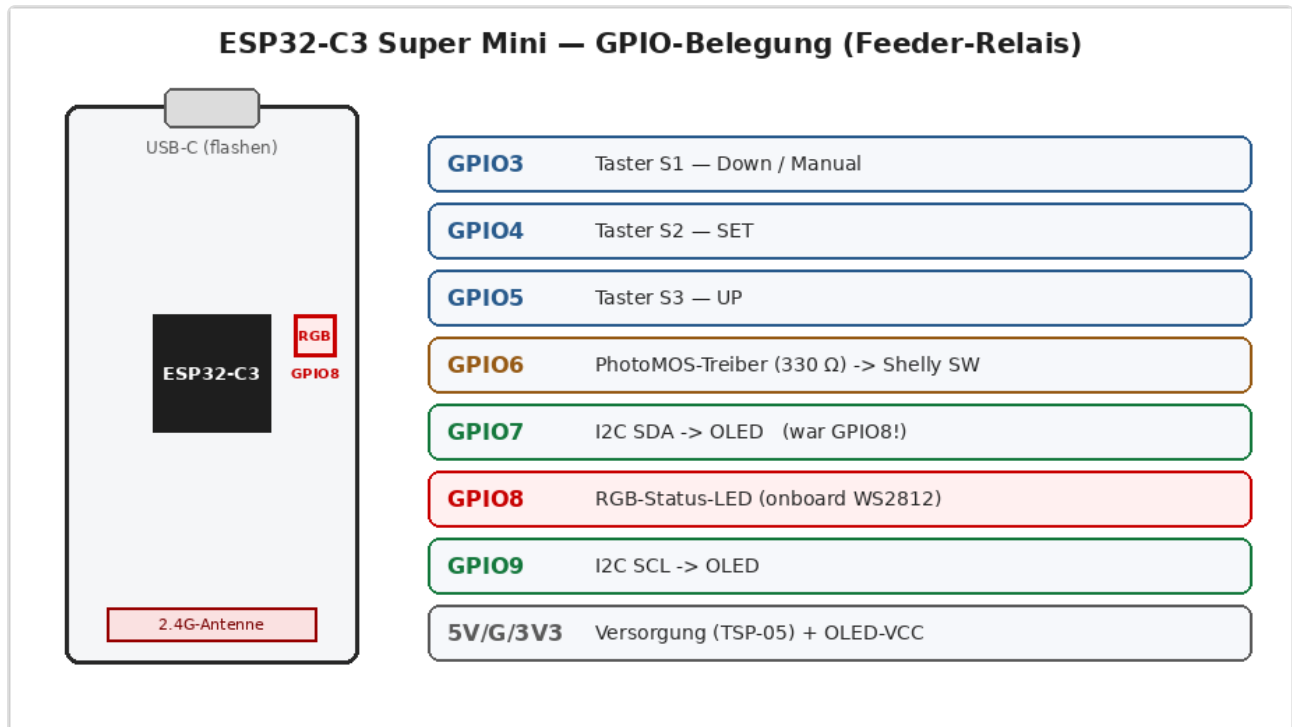
5.1 Overview of the nets (connections)

Net name	carries	connects
L_IN	230 V	mains terminal J1.L → fuse F1
L_F	230 V (fused)	F1 → varistor, TSP-05 AC, PhotoMOS OUT1, Shelly L
N	230 V neutral	J1.N → varistor, TSP-05 AC, Shelly N, load J3
SW_SHELLY	230 V switched	PhotoMOS OUT2 → Shelly SW
O_LAST	230 V switched	Shelly O → load output J3
+5V	low voltage (SELV)	TSP-05 +Vo → ESP 5V, C1, C2
GND	low voltage (SELV)	TSP-05 −Vo → ESP GND, buttons, PhotoMOS K, OLED
+3V3	low voltage (SELV)	ESP 3V3 output → OLED VCC
BTN1/2/3	signal	buttons → GPIO3/4/5 (internal pull-ups)
PMOS_DRV → PMOS_A	signal	GPIO6 → R1 330 Ω → PhotoMOS anode
SDA / SCL	I2C	GPIO7 / GPIO9 → OLED

Net name	carries	connects
RGB-LED	signal	GPIO8 → onboard WS2812 (status LED, dimmed)

5.2 ESP32-C3 Super Mini: board and pin assignment

Module variant "ESP32-C3FN4". An overview of the pins we use:



Components on the module: **1** USB Type-C · **2** BOOT button · **3** RESET button · **4** LDO CAT6219 (3.3 V, 500 mA) · **5** ESP32-C3FN4 · **6** 2.4 GHz antenna · **7 RGB module (WS2812, GPIO8)** · **8** antenna connector (u.FL).

Pin	Function	Note
5V / G	supply	from the TSP-05
3V3	output of the onboard LDO	supplies the OLED
GPIO3	button T1	pull-up internal, button to GND
GPIO4	button T2	ditto
GPIO5	button T3	ditto
GPIO6	PhotoMOS driver	high = Shelly on
GPIO7	SDA (I2C)	OLED — instead of GPIO8 , which is where the RGB LED sits
GPIO8	RGB LED (WS2812)	onboard status LED, dimmed
GPIO9	SCL (I2C)	OLED

Deliberately avoided: GPIO2/8/9 as buttons (strapping pins). **GPIO8 carries the onboard RGB LED (WS2812)** — which is why I2C **SDA is on GPIO7** (not GPIO8), otherwise the I2C traffic would drive the LED brightly (heat). SCL stays on GPIO9 (boot-compatible with an I2C pull-up).

5.3 Shelly 1PM Mini Gen4

- Terminals: **SW** (switch input), **O** (load output), **L**, **N**.

- Supply 110–240 V~, relay max. 8 A / 2000 W @ 240 V.
- Configuration after commissioning: Input Mode = Switch (Follow), so that the relay follows the SW signal 1:1.
- Mounting: lying flat on the mounting area, VHB adhesive tape or 3D clip. Four short wires (e.g. 0.75 mm², 6–7 mm stripped) from the screw terminals to the J2 pads.

5.4 Power supply TSP-05

100–240 V~ → 5 V DC / 3 W (= 600 mA). Sufficient for the ESP Wi-Fi peaks (~350 mA) plus the OLED with reserve. **Open item: check the pin spacing with calipers against the KiCad footprint Converter_ACDC_HiLink_HLK-PMxx** (the clone is usually pin-compatible — but at 230 V, assume nothing).

5.5 Buttons

The three 6×6 short-travel buttons (1.5 mm travel) are actuated by plungers in the front panel — their position is therefore **non-negotiable** (place them to within ±0.2 mm). Pin pattern per button: 6.9 mm × 4.4 mm.

5.6 OLED — plug-in mounting ("stack")

The OLED is **not soldered**, but **plugged onto J4**: socket header 1×4 on the board, pin header on the OLED facing backward, the display face rests against the glass pane (with a thin foam pad).

- Assignment of J4 from top to bottom: **GND, VCC (3V3), SCL, SDA** — identical to the labeling of the existing module.
- **Required stack height** = 16.5 mm – glass inset – module thickness (≈ 2.8 mm) ≈ **13–14 mm**. Realized for example with an 11 mm socket header + 2.5 mm pin header collar. Socket headers are available in 8.5/11/13/15 mm.
- **Measure on the real module before freezing** (manufacturers vary by ±1 mm): 1. distance module left edge → pin row (assumption 2.5 mm), 2. distance module left edge → center of the active area (assumption 23.5 mm), 3. glass inset (front inner face → underside of glass). In practice: lay the module in the frame, display centered in the window, measure the pin holes relative to two boss screws → that is J4.

6. Firmware (ESPHome)

File: `firmware/timer-relais-c3.yaml` (device name **feeder-relais**, mDNS `feeder-relais.local`). Contains everything: Wi-Fi with fallback hotspot ("Feeder-Relais Setup", captive portal), **NTP clock** (`de.pool.ntp.org`), mobile web app + JSON API (Ch. 6.4), the three persistent times, button logic with debounce/long press/menu (Ch. 6.5) and an OLED with Wi-Fi status, clock or countdown.

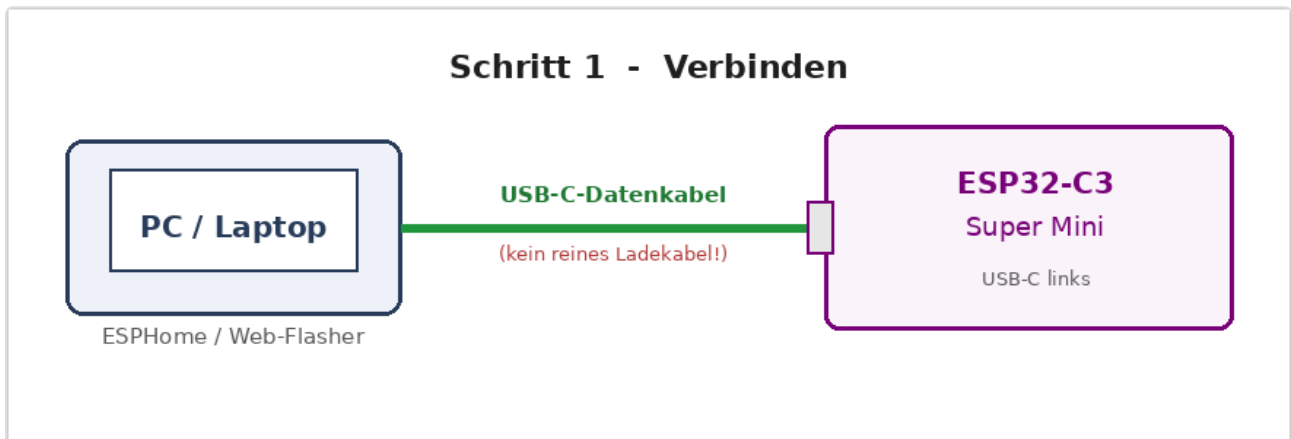
6.1 First flashing — step by step (for beginners)

The **very first time**, the firmware is put onto the ESP32-C3 **via a USB cable**. All later updates then run wirelessly (Ch. 6.6).

This is what you need:

- the **ESP32-C3 Super Mini**,
- a **USB-C data cable** — caution: many cheap cables can only charge; with those the chip is **not detected**!

- a PC with **Google Chrome** or **Microsoft Edge** (browser route B) — or ESPHome on the PC (route A).



Ready-made images are available — so you don't have to compile anything yourself:

- in the repo under `firmware/build/` (always the current state):
- `feeder-relais.factory.bin` — full image for the **first flash** (address 0x0)
- `feeder-relais.ota.bin` — app image for **OTA/web update** (Ch. 6.6)
- as a **GitHub release** (versioned file name, e.g. `feeder-relais-v0.0.1.factory.bin`) in the Releases section of the repository.

Build it yourself: `./firmware/build_images.sh` (compiles and updates `firmware/build/`). Otherwise the raw files are located under `firmware/.esphome/build/feeder-relais/.pioenvs/feeder-relais/`.

The images contain NO Wi-Fi credentials. Setup runs after flashing via the setup hotspot (Ch. 6.2). This way no foreign Wi-Fi ends up in the delivered product, and the configured data survives OTA updates (fixed storage key).

Route A — with ESPHome (recommended, one command)

A `secrets.yaml` is **not** needed — no Wi-Fi data is compiled in.

1. Plug in the ESP via **USB-C**.
2. Run in the project folder: `esphome run firmware/timer-relais-c3.yaml` ESPHome compiles, asks for the **serial port** (Linux e.g. `/dev/ttyACM0`, Windows a `COMx`) and flashes.
3. After the restart the ESP opens the setup hotspot → continue with Ch. 6.2.

Linux tip: On "Permission denied" for `/dev/ttyACM0`, add your user to the group `dialout`:
`sudo usermod -aG dialout $USER` (log in again).

Route B — in the browser, without installation (esptool-js)

For the ready-made `firmware.factory.bin` entirely without ESPHome:

Ablauf - Erstes Flashen ueber den Browser (esptool-js)

- 1 Chrome oder Edge oeffnen**
espressif.github.io/esptool-js
- 2 ESP per USB-C anstecken**
USB-C-Datenkabel verwenden
- 3 "Connect" -> Port waehlen**
ESP32-C3 als serieller Port
- 4 Flash Address 0 + factory.bin**
Datei waehlen, dann "Program"
- 5 WLAN einrichten**
AP "Feeder-Relais Setup" -> feeder-relais.local

1. In **Chrome/Edge** open the page <https://espressif.github.io/esptool-js/>.
2. Plug in the ESP via a **USB-C data cable**.
3. Leave the baud rate at 115200, click **Connect** and select the ESP's serial port in the window (often called "USB JTAG/serial debug unit" or a COMx).
4. Enter 0 at **Flash Address**, use **Choose File** to select the `firmware.factory.bin`, then **Program**. Wait until "Hard resetting..." appears.
5. Continue with the Wi-Fi setup (Ch. 6.2).

If "Connect" does not work: force **boot mode** — hold **BOOT**, briefly tap **RESET**, release **BOOT**, then "Connect" again.

6.2 After flashing: set up Wi-Fi

The firmware **deliberately brings no** Wi-Fi credentials with it — after flashing the ESP starts its own **setup hotspot**:

1. On your phone/PC connect to the Wi-Fi **"Feeder-Relais Setup"** (password `feeder1234`).
2. A **captive portal** opens (otherwise open <http://192.168.4.1>).
3. Select your home Wi-Fi, enter the password, save — the ESP restarts and connects.
4. From now on reachable at <http://feeder-relais.local>.

Persistence & factory reset:

- The entered Wi-Fi data is stored under a **fixed storage key** and is **preserved across OTA/web updates** (Ch. 6.6) — set up once, then never again.
- For a **factory reset** (deleting the old Wi-Fi data), flash the **factory image with flash erase**: tick "Erase all flash" in the web flasher, or run `esptool.py erase_flash` before writing. A pure OTA update does **not** delete the data.

6.3 Troubleshooting when flashing

Symptom	Cause / solution
ESP is not detected at all	charge cable instead of data cable → different USB-C cable, different USB port
No port in the browser	use Chrome/Edge (WebSerial); on Windows possibly the CH340/CP210x driver; a C3 with native USB usually needs none
"Connect" fails	boot mode : hold BOOT + tap RESET + release BOOT
Linux "Permission denied"	user in group dialout (sudo usermod -aG dialout \$USER)
Nolocal after the flash	first set up Wi-Fi via the setup hotspot (Ch. 6.2); mDNS takes a moment

After the first flash all updates go **wirelessly** (Ch. 6.6) — the USB socket may well be poorly accessible in the installed state.

6.4 Web operation and JSON API (port 80)

`http://feeder-relais.local` opens a mobile web app with five tabs:

- **Start**: three large buttons (trigger T1/T2/T3 with their configured time), live countdown and **Stop**.
- **Times (settings)**: at the very top the **language selection** (Deutsch/English/Français), below it the three timer times (1–600 s), stored persistently.
- **Network (configurable)**: Wi-Fi (SSID/password), IP mode **DHCP/static** (+ IP/gateway/mask/DNS), **NTP server**, a **hostname** (default `feeder-relais`), a switch for **Wi-Fi roaming (802.11k/v)** and a **restart** button.
- **Status**: firmware, uptime, free memory, Wi-Fi, SSID with **channel · colored signal bar · signal** (dBm; 1 bar red, 2 yellow, 3–4 green), IP/MAC, reset reason, relay.
- **Service**: live **log** (display level ERROR/WARN/INFO/DEBUG selectable, can be enabled), **firmware update** (.bin upload, Ch. 6.6) and **restart**.

Language (multilingual interface): The web app and the OLED speak **German, English and French**. The selection is **at the very top of the Times tab** and applies **device-wide** (web and OLED), stored in `g_netcfg.lang` (persistent, default German). The web texts are stored as a JS dictionary in the page (switching without reload); language-dependent status values (Wi-Fi state, reset reason) come from the firmware as neutral codes and are translated in the browser. The OLED translates weekdays, the status (Idle/Feed) and the menu titles (deliberately without accents because of the display's character set). The manual itself is additionally available in [English](#) and [French](#) (the German version is authoritative).

In the **header** are the **hostname** and a **status dot**: green = all is well, yellow = a timer is running, red = fault in the idle state (e.g. OLED not reachable or no Wi-Fi). The **same traffic light** is shown on the device by the dimmed **onboard RGB LED** (WS2812 on GPIO8, see Ch. 5.2).

Technically, `firmware/timer_web.h` (with `firmware/net_config.h` for the persistent network config) serves this page as its own `AsyncWebHandler` on `web_server_base` (the native ESPHome web UI is disabled). The same **JSON API** is also used for integration (e.g. ioBroker) — parameters as a query string, method GET **or** POST:

Endpoint	Effect
GET /api/status	

Endpoint	Effect
	JSON with all values: active, remaining, relay, last, times[3], host, ip, ssid, rssi, mac, ap, fw, uptime, heap, wifi, reset, lang (wifi/reset are language-neutral codes that the web JS translates)
POST /api/trigger?button=N	trigger button N (1-3) (uses its configured time)
POST /api/trigger?seconds=N	switch on ad hoc for N seconds
POST /api/stop	switch off immediately
POST /api/config?time1=A&time2=B&time3=C	set times (each 1-600 s, persistent) — fields can also be set individually
GET /api/net	read network config: static, ip, gw, sn, dns, ntp, hostname, roaming, lang
POST /api/net?static=0\ 1&ip=&gw=&sn=&dns=&ntp=&host=&roaming=0\ 1&lang=de\ en\ fr	save network config (fields individually, persistent)
POST /api/wifi?ssid=&pw=	set Wi-Fi credentials (reconnects)
POST /api/reconnect	cleanly reconnect Wi-Fi (applies e.g. changed roaming immediately)
POST /api/reboot	restart the device
GET /api/log?level=N&since=M	log ring buffer as JSON (lines with level ≤ N, seq > M); level 1=ERROR...5=DEBUG

Applying the network config: The values are stored in the ESPHome preferences (flash). The **NTP server** takes effect at the next sync (`esp_sntp_setservername`). The **static IP** is set at `wifi.on_connect` via the ESP-IDF `netif` and is therefore active after a **restart** (DHCP is the default). **Wi-Fi** uses ESPHome's own `save_wifi_sta()`; initial access always via the captive portal. The **hostname** is switched at runtime: mDNS (`...local`) immediately via `mdns_hostname_set()`; the DHCP name (in the router) by a **restart of the DHCP client** (`esp_netif_dhcpc_stop/start` → fresh DISCOVER with option 12), i.e. also without a reboot. The name also appears in the **header** of the web app and in the browser tab. It is reduced to a valid DNS label (a-z, 0-9, "-").

Wi-Fi roaming (802.11k/v): The capability is compiled in permanently (`enable_btm/enable_rrm` in the YAML → `CONFIG_WPA_11KV_SUPPORT` in the `wpa_supplicant`); the **on/off** is toggled at runtime by the web switch via `g_netcfg.roaming` (persistent). On sets in the STA config the bits 802.11v **BTM** (`set_btm`, the AP/router can specifically re-book the device onto the stronger AP) and 802.11k **RRM** (`set_rrm`, neighbor AP lists) and turns off ESPHome's own scan roaming (the driver takes over); off does the reverse (`set_post_connect_roaming(true)`). Only sensible with **multiple access points with the same SSID** (UniFi/mesh) and when these support 802.11k/v. The bits only go into the STA config at the **next (re)connect**; for that there is the button "**Reconnect now**" in the same card (`POST /api/reconnect` → `wifi.disable + enable`, which briefly disconnects the Wi-Fi), otherwise it takes effect at the next restart. Default: **off**.

Example ioBroker (set the time for button 1 to 8 s): `POST http://feeder-relais.local/api/config?time1=8`.

6.5 Operation via the three buttons and OLED

The buttons are labeled on the enclosure: **S1 = Down/Manual**, **S2 = SET**, **S3 = UP** (GPIO3/4/5).

Normal state:

- **short S1 / S2 / S3** → triggers timer 1 / 2 / 3 with the configured time (switches the Shelly on until the time expires).
- **long UP (S3) ≥ 1.2 s** → stops all timers and switches the Shelly **off**.
- **long SET (S2) ≥ 3 s** → opens the **info menu**.

Info menu (view only — configuration is done via the web app):

- **S1** pages forward, **S3** backward; **short SET** or **10 s without a press** closes the menu again.
- Pages: 1) Wi-Fi (SSID + signal) · 2) IP address · 3) time + NTP · 4) system (firmware + uptime).

OLED display (128 × 32):

- **Top:** Wi-Fi signal bars (from RSSI) on the left, the large **time of day** (HH:MM) or, while a timer is running, the **countdown** (e.g. "10 s") in the center, status **Idle/Feed** on the right; in the menu instead the page counter (e.g. "2/4").
- **Bottom:** on the left the **date** (e.g. "Do 16.07.2026"), on the right the **free memory** (kB).

The time base comes via NTP; until the first synchronization the clock shows "--:--". Time zone Europe/Berlin. Weekday, status (Idle/Feed) and the menu titles follow the **configured language** (Ch. 6.4).

6.6 Firmware updates (OTA)

After the first USB flash, **two wireless update routes** are set up:

- **Network OTA (ESPHome):** `esphome run firmware/timer-relais-c3.yaml` updates over Wi-Fi (port 3232) — USB no longer needed. Configured via `ota: platform: esphome`.
- **Web upload:** In the **Service** tab, upload the compiled `firmware.bin` (or directly `POST /update` on port 80). Configured via `ota: platform: web_server` — runs over `web_server_base`, the native ESPHome web UI is **not** needed for it. The device restarts automatically after the update.

The `firmware.bin` is created by `esphome compile ...` and is located under `.esphome/build/feeder-relais/.pioenvs/feeder-relais/firmware.bin`. The captive portal can also install updates.

6.7 Service log / debugging

The **Service** tab shows the last log lines (ring buffer, 40 lines) filtered by the selected level (ERROR/WARN/INFO/DEBUG) when "live display" is enabled. Technique: `logger: on_message:` writes each line into `firmware/log_ring.h`; the web endpoint `GET /api/log?level=&since=` delivers them as JSON. The logger runs at `level: DEBUG` (for VERBOSE, raise the level in the YAML). Complete logs additionally via `esphome logs ...` (UART/network).

7. KiCad — step by step (also for beginners)

Tested against KiCad 8/9/10; menu names may differ slightly.

7.1 Create a project and import the schematic

1. Start KiCad → **File** → **New Project** → name `timer_ersatzplatine`, folder e.g. `hardware/kicad/`.
2. KiCad creates an empty `timer_ersatzplatine.kicad_sch`. **Replace this file in the explorer with our file** (same name!): copy in `hardware/timer_ersatzplatine.kicad_sch`.

3. Open the project, start the schematic editor. KiCad may convert the file to the current format — that is normal, just save.
4. Run **Tools → Electrical Rules Check (ERC)**. Warnings about "unconnected power pins" are known and non-critical (our symbols deliberately have no power-pin types); optionally silence them with PWR_FLAG symbols on +5V and GND.

7.2 Check/assign footprints

Tools → Assign Footprints. Pre-assigned are:

Ref	Footprint	Action
SW1-3	Button_Switch_THT:SW_PUSH_6mm	check (pin pattern 6.9 × 4.4)
PS1	Converter_ACDC:Converter_ACDC_HiLink_HLK-PMxx	check against the TSP-05 measurement!
U2	Package_S0:SOP-4_4.4x2.6mm_P1.27mm	for the DIP variant (AQY216 DIP): Package_DIP:DIP-4_W7.62mm
F1	Fuse:Fuse_5x20mm_Horizontal_ReferenceFuseHolder	or the chosen holder
RV1	Varistor:RV_Disc_D12mm_W3.9mm_P7.5mm	S14K275 fits
R1, C1, C2	standard THT	ok
J1-J3	pin header/terminal	J2 alternatively as 4 large solder pads
J4	Connector_PinSocket_2.54mm:PinSocket_1x04_P2.54mm_Vertical	socket header!
U1	— (empty)	create yourself , see 7.3

7.3 Create a footprint for the ESP32-C3 Super Mini

1. Open the footprint editor → new library `timer_project.pretty` (in the project folder, table "project-specific").
2. New footprint `ESP32-C3_SuperMini`: two pad rows 1×8, pitch 2.54, row spacing 15.24 mm (= 6 × 2.54), pads ø1.7/hole 1.0. Outline 18 × 24 mm on F.Fab, mark the USB side.
3. Assign pad numbers according to the pinout image (left row 5V, G, 3.3, 4, 3, 2, 1, 0 / right row 5, 6, 7, 8, 9, 10, 20, 21) and assign them in the schematic to symbol U1 (the symbol pin names 5V/G/3V3/3/4/5/6/8/9 must point to the correct pad numbers).

7.4 Generate the PCB from the schematic

Schematic editor → **Tools → Update PCB from Schematic**. All footprints land as a heap next to the (still empty) board.

7.5 Import the contour and references

1. PCB editor → **File → Import → Graphics** → `hardware/platine_original_geometrie.dxf`.
2. The unit is automatically set to **mm** thanks to the DXF header; import first onto layer **User.Drawings**, position (0,0), scale 1.
3. **Dimension check**: measure T1→T2 with the measuring tool — it must give **25.6 mm**. If it is correct, continue; otherwise check the import settings.

4. Select the outer contour (**one closed line** — the 6 edge-open boss slots are already part of the contour, no longer separate circles) → Properties → change layer to **Edge.Cuts**. Everything else (button crosses, mounting areas, window, keepouts, texts) stays on user layers; optionally copy the mounting area frames including their labels additionally onto **F.Silkscreen** (helps with placement).
5. Set the 4 mounting holes as footprints: `MountingHole:MountingHole_3.2mm` exactly on the green crosses (position via **E** → type in the coordinates from Ch. 4.3).

7.6 Place and route

1. **Buttons first:** SW1–SW3 exactly on (13.2/12.3), (38.8/12.2), (64.1/12.0) — enter the position numerically, do not drag.
2. J4 on the pin crosses (first pin = GND at x 15.0 / y 27.84).
3. PS1/TSP on the mounting-area rectangle (AC pins toward the left edge), U1/ESP on its area (USB toward the top edge), J2 pads at the edge of the Shelly mounting area, J1 + F1 + RV1 in the 230 V corner at the bottom left, U2/PhotoMOS between the ESP area and the Shelly pads, R1 next to it, C1/C2 on the 5 V line near U1.
4. Create **net classes** (File → Board Setup → Net Classes): - HV (nets L_IN, L_F, N, SW_SHELLY, O_LAST): trace ≥ 0.8 mm (the supply path of the Shelly output O ≥ 2 mm at 8 A load!), clearance within HV ≥ 1.0 mm. - Standard (low voltage): 0.25/0.2 mm.
5. **The golden rule:** between every HV net and every low-voltage net ≥ 6.0 mm distance — most easily via a custom rule (Board Setup → Custom Rules): (version 1) (rule HV_zu_LV (condition "A.NetClass == 'HV' && B.NetClass != 'HV'") (constraint clearance (min 6.0mm))) The only permitted "approach" is the PhotoMOS itself — its housing is the separation point (pins 1/2 = LV, pins 3/4 = HV).
6. No ground planes/zones in the HV area; no foreign traces under the TSP module.
7. Get **DRC** (Design Rules Check) free of errors.

7.7 Manufacturing data and ordering

1. **File → Fabrication Outputs → Gerbers** (standard layers) + drill files.
2. 3D view (Alt+3) as a final visual inspection.
3. Manufacturer (JLCPCB/PCBWay/Aisler): 2 layers, 1.6 mm FR4, lead-free HASL or ENIG, color does not matter. The outer contour including the 6 edge-open boss slots comes automatically from Edge.Cuts.
4. **Before submitting:** print the board 1:1 on paper (File → Plot → PDF, scale 1:1), cut it out and test-fit it in the enclosure!

8. Assembly and commissioning

1. **Place the small parts** (order flat → tall): R1, C2, U2, buttons (exactly!), J4 socket header, C1, F1 holder, RV1, terminals.
2. **Prepare the ESP32-C3:** flash via USB (Ch. 6), test the Wi-Fi connection, then solder it onto the board (pin headers or directly).
3. **USB-only test:** power the ESP via USB (no mains!) — the buttons must start a countdown, the OLED plugged onto J4 must display, GPIO6 must switch (multimeter/LED on the PhotoMOS input).
4. **Solder in the TSP-05,** glue the Shelly onto the mounting area with VHB, four wires to J2 (L, N, SW, O), a wire to the load output.

5. **Visual inspection + continuity check:** no short between L/N/PE, ≥ 6 mm distances maintained, no solder splashes.
6. **First mains test:** enclosure closed, switch on via PRCD/RCD. The OLED shows clock/status → set up the Shelly via button/app (Wi-Fi, Input Mode "Switch") → button test with load.
7. Set the times as desired via `http://feeder-relais.local`.

9. File overview

File	Content	Regenerable?
<code>hardware/platine_template.py</code>	source of truth for all geometry; parametric	source
<code>hardware/platine_original_geometrie.dxf</code>	DXF generated from it for KiCad	yes: python3 <code>platine_template.py</code> (needs <code>pip install ezdxf</code>)
<code>hardware/platine_vorschau.png</code>	render of the DXF	yes (render script see CLAUDE.md)
<code>hardware/timer_ersatzplatine.kicad_sch</code>	complete schematic (KiCad 8+)	hand-maintained
<code>firmware/timer-relais-c3.yaml</code>	ESPHome configuration	hand-maintained
<code>firmware/timer_web.h</code>	mobile web app + JSON API (C++ handler on <code>web_server_base</code>)	hand-maintained
<code>firmware/net_config.h</code>	persistent network config (IP mode, static IP, NTP, hostname, 802.11k/v roaming)	hand-maintained
<code>firmware/log_ring.h</code>	log/debug ring buffer for the Service tab (<code>/api/log</code>)	hand-maintained
<code>firmware/build/*.bin</code>	ready-made flash images (factory + ota), current state	generated
<code>firmware/build_images.sh</code>	compiles and updates <code>firmware/build/</code>	source
<code>docs/DOKUMENTATION.md</code>	this document (German source , authoritative)	hand-maintained
<code>docs/DOKUMENTATION.en.md</code> , <code>docs/DOKUMENTATION.fr.md</code>	manual in English/French (translation, updated per release)	hand-maintained
<code>docs/PROJEKTPLAN.md</code>	phases, status, checklists (German only)	hand-maintained
<code>CLAUDE.md</code>	working instructions for Claude Code	hand-maintained
<code>docs/DOKUMENTATION.pdf</code> , <code>docs/PROJEKTPLAN.pdf</code>	PDF versions (mandatory with every change)	yes: python3 <code>tools/md2pdf.py</code> <code>docs/*.md</code>
<code>tools/md2pdf.py</code>	Markdown→PDF generator (embeds images)	source
<code>docs/img/*.png</code>	diagrams of the flashing guide (Ch. 6.1)	generated (PIL)

10. Change history of the geometry template

Version	Change
v1	first coordinates from perspective-corrected photos ($\pm 1\text{--}2$ mm)
v2	scanner measurement: 76.7×66.1 , buttons/holes/cutout precise (solder-side scan)
v3	enclosure fit 105.3×79.4 , 6 boss cutouts $\varnothing 10$, old cutout omitted (= B5)
v4	correction: no recessing possible (only 1.8 mm below) — window removed, Shelly on the mounting area at the right, 4th hole back, rib keepouts
v5	TSP-05 planned instead of HLK (bottom left), ESP to the top left
v6	OLED frame-mounted: window area freed up, J4 connection
v7	labeled placeholders (DXF texts)
v8	OLED plug-in mounting: J4 constructed vertically on the window centering
v9	boss positions measured directly (Silvio 13.07.: pitch 45, row spacing 70, upper line 6 mm above the top edge, left column 6 mm beside the left edge $\rightarrow x -6/39/84, y -6/+64$); 6 boss cutouts changed from $\varnothing 10$ holes to edge-open U-slots (more tolerant during insertion); corner radii at the outer slots locally reduced to 3.2/2.1; contour remains a closed Edge.Cuts loop (verified)